

UNIT : 3

GIGABYTE™



Probability

- Probability manages uncertainty in noisy data.
- Values range from zero to one.
- Confidence scores replace rigid logical rules.
- Distributions distinguish normal patterns from noise.
- Probabilities define errors within loss functions.
- Randomness becomes reasoned logic through math.



Types of probabilities

1. Joint probability:

- Joint probability calculates multiple events occurring together.
- Example: $P(A \cap B)$ represents both events happening.

2. Conditional probability:

- Conditional probability calculates event likelihood given prior.
- Node tables show parent-child state combinations.

If A and B are dependent events $p(B/A) = P(A \text{ and } B) / P(A)$

If A and B are independent events : $P(B/A) = P(B)$

Bayes' theorem

- Probability measures uncertainty in events.
- Bayes' Theorem updates probabilities with evidence.
- calculates conditional probability based on prior.
- Bayes' Theorem combines prior and likelihood.
- It refines beliefs as new data arrives.
- Widely used in machine learning models.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Diagram illustrating the components of Bayes' Theorem:

- $P(B|A)$: Probability of B occurring given evidence A has already occurred
- $P(A)$: Probability of A occurring
- $P(A|B)$: Probability of A occurring given evidence B has already occurred
- $P(B)$: Probability of B occurring

BAYES THEOREM

- **Prior probability:** initial likelihood without evidence.
- **Posterior probability:** updated likelihood with evidence.

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)}$$

- $P(h)$ = prior probability of hypothesis h
- $P(D)$ = prior probability of training data D
- $P(h|D)$ = probability of h given D
- $P(D|h)$ = probability of D given h



BAYES THEOREM

- **Prior Probability ($P(A)$):** Initial belief about event A before considering new evidence.
- **Likelihood ($P(B|A)$):** Probability of evidence B occurring if A is true.
- **Evidence ($P(B)$):** Overall probability of observing evidence B, used for normalization.
- **Posterior Probability ($P(A|B)$):** Updated belief about A after considering evidence B.



Example of Spam Email:

- $P(S)=0.2$ (20% of emails are spam)
- $P(H)=0.8$ (80% of emails are not spam)
- $P(W | S)=0.6$ (60% of spam emails contain the word "win")
- $P(W | H)=0.1$ (10% of non-spam emails contain the word "win")

Calculate $P(S|W)$.

Probability & Bayes' theorem

Bayes Theorem

Find the Probabaility of Pet in the Box given that the Pet is quiet.



Bayes Theorem

Probability that the Pet is Quiet Cat is given by



$$P(\text{Cat} \mid \text{Quiet}) = \frac{P(\text{Quiet Cat}) \times P(\text{Cat})}{P(\text{Quiet})}$$

Bayes Theorem

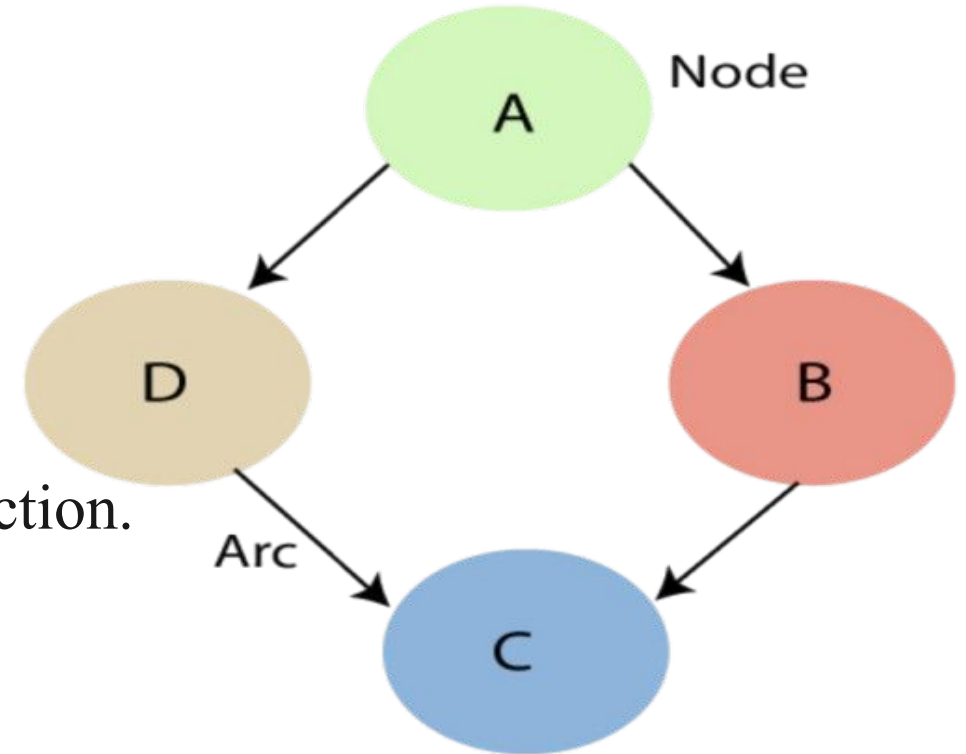
Probability that the Pet is Quiet Dog is given by



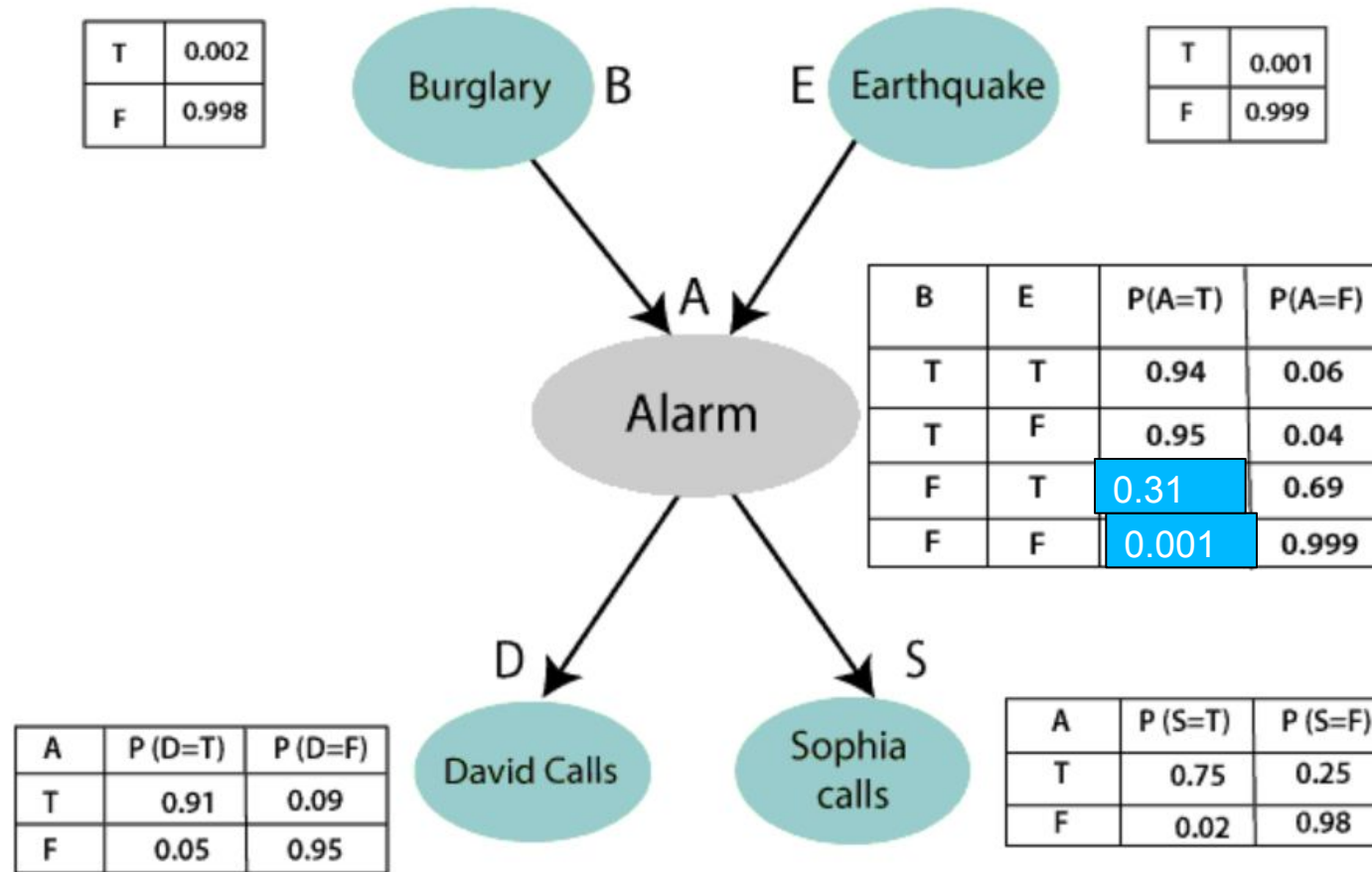
$$P(\text{Dog} \mid \text{Quiet}) = \frac{P(\text{Quiet Dog}) \times P(\text{Dog})}{P(\text{Quiet})}$$

Bayesian networks

- Bayesian networks model uncertainty in AI.
- They represent dependencies using directed graphs.
- Tables define variable relationships.
- Used for prediction and decision-making tasks.
- AI applications include diagnostics and anomaly detection.
- Helps AI reason under uncertain knowledge.



Bayesian networks



$$P(B = T \mid D = T, S = T)$$



Statistics

- Statistics in AI is used to collect, analyze, and understand data.
- It helps AI systems find patterns and relationships in data.
- Statistics is used to calculate probability and uncertainty in AI models.
- AI uses statistical methods to make predictions and decisions.
- Concepts like mean, variance, probability, and distribution are important in AI.
- Statistics help improve the accuracy and performance of machine learning models.



Machine learning

- 1 Machine learning is a subset of artificial intelligence.
- 2 Predicting accurate outcomes is the strength of machine learning algorithms.
- 3 The models in machine learning are designed to conclude the most accurate predictions possible.
- 4 Machine learning is all about outcomes.



Statistics

- 1 Statistics is a field of mathematics that studies data through various techniques.
- 2 The statistical models are intended for inference about the connections between the variables.
- 3 Many statistical models make predictions, but they are not accurate enough.
- 4 Statistics is all about finding relationships between variables and their significance.



Types of Statistics

- **Descriptive Statistics**—summarizes and describes the main features of a dataset using measures like mean, median, and graphs.
- **Inferential Statistics**—deals with making predictions and inferences about a population based on a sample of data. It involves using hypothesis testing, confidence intervals, and regression analysis to draw conclusions about the data.



Descriptive Statistics	Inferential Statistics
1. Works with a smaller data set	1. Works with a large data set
2. Process is simpler to do	2. Process is more complex as we have to decide on the best sampling techniques
3. Results obtained represent the entire data set	3. Results obtained represent a portion of the population, but can be used to deduce information about the entire population with some amount of uncertainty
4. Error involved is usually less	4. Error involved is usually more

Descriptive Statistics

- **Mean:**

- The average of all values in a dataset.
- Example: Data = 2, 4, 6 $\rightarrow$ Mean = 4

- **Median:**

- The middle value when data is arranged in order.
- Example: Data = 1, 3, 5 $\rightarrow$ Median = 3

- **Mode:**

- The value that appears most frequently in data.
- Example: Data = 2, 2, 4, 5 $\rightarrow$ Mode = 2



Descriptive Statistics

```
import numpy as np
import pandas as pd

data = np.array([1, 2, 3, 4, 5])
df = pd.DataFrame(data, columns=["Values"])
print(df.describe())
```

	Values
count	5.000000
mean	3.000000
std	1.581139
min	1.000000
25%	2.000000
50%	3.000000
75%	4.000000
max	5.000000



Descriptive Statistics

```
import numpy as np
import pandas as pd
# create a sample salary table
salary = pd.DataFrame({
    'employee_id': ['001', '002', '003', '004', '005', '006', '007',
                    '008', '009', '010'],
    'salary': [50000, 65000, 55000, 45000, 70000, 60000, 55000, 45000,
               80000, 70000]
})

# calculate mean
mean_salary = np.mean(salary['salary'])
print('Mean salary:', mean_salary)

# calculate median
median_salary = np.median(salary['salary'])
print('Median salary:', median_salary)

# calculate mode
mode_salary = salary['salary'].mode()[0]
print('Mode salary:', mode_salary)
```



Variance

- **Variance** measures how far each data value is spread from the **mean (average)**.

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

Where:

- x = each value
- μ = mean
- N = total number of values

Standard Deviation (SD)

- **Standard Deviation** is the **square root of variance**. It shows how much data is spread from the mean in the **same units as the data**.

$$SD = \sqrt{Variance}$$

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

```
import numpy as np

data = np.array([1, 2, 3, 4, 5, 6])
std_dev = np.std(data)

print('Standard deviation:', std_dev)
```



Bias, Variance, and Model Error

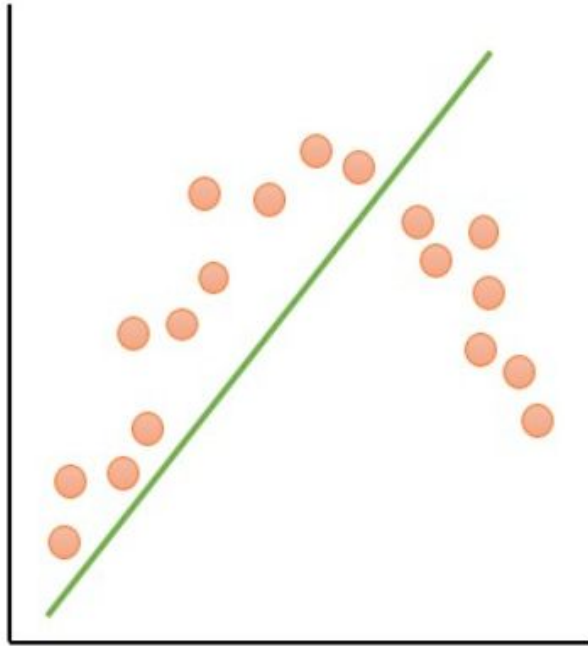
- The performance of a machine learning model is fundamentally governed by the bias–variance tradeoff.
- **Bias:** Error caused by a model being too simple and missing the true pattern.
- **Variance:** Error caused by a model being too complex and reacting too much to training data.
- **Irreducible Error:** Error caused by noise or randomness in the data that cannot be eliminated.



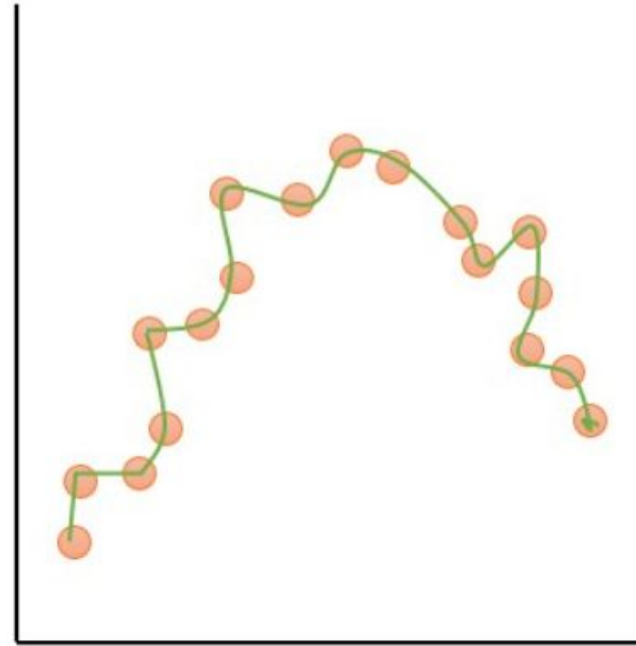
Bias

- Bias is systematic prediction error.
 - **High bias** means overly simple model.
 - **High bias** causes underfitting problems.
 - **Low bias** fits training data well.
 - **Low bias** often increases variance.
- Overly complex models reduce bias.

High Bias, Underfitting



Low Bias, Overfitting



Orange points are actual data, and the green line is the model trying to predict the data.



Variance

- Variance measures how spread predictions are.
- It shows how predictions change.
 - **High variance** means predictions vary greatly.
 - **High variance** causes overfitting problem.
 - **Low variance** means stable predictions.
 - **Low variance** can cause underfitting.



- Green dot is the **mean (average)** of the data.



Feature	Bias	Variance
Meaning	Error from wrong assumptions (too simple)	Error from sensitivity to data (too complex)
Model type	Simple model	Complex model
Effect on training data	High error	Low error
Effect on test data	High error	High error
Problem caused	Underfitting	Overfitting



Percentiles

- Percentiles describe data distribution in datasets.
- They show percentages below certain values.
- The 25th percentile equals the first quartile value.
- The 75th percentile equals the third quartile value.
- Used for preprocessing and exploratory analysis.
- NumPy and Pandas calculate percentiles easily.

```
import numpy as np

data = np.array([1, 2, 3, 4, 5])
p25 = np.percentile(data, 25)
p75 = np.percentile(data, 75)
print('25th percentile:', p25)
print('75th percentile:', p75)
```



Types of data distribution

1. Normal Distribution
2. Skewed Distribution
3. Uniform Distribution



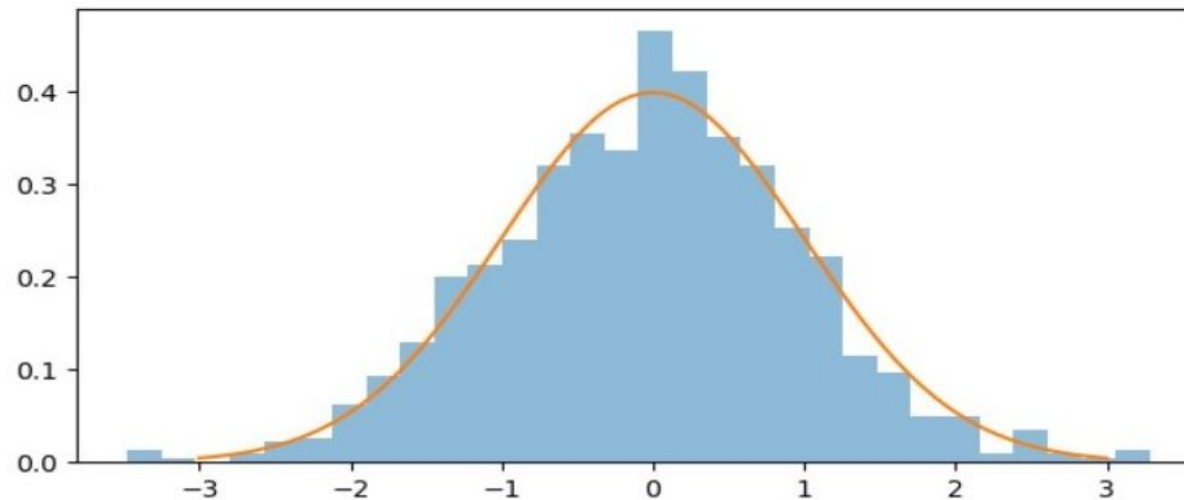
Normal Distribution

A **normal distribution** is a way to describe how values are spread out in data. It is also called a **Gaussian distribution**.

It looks like a **bell-shaped curve**:

- The **middle** of the curve is the **mean** (μ) — the average value.
- The **spread** of the curve depends on the **standard deviation** (σ) — how far values are from the mean.
- It is **symmetric**, meaning the left side and right side are mirror images.

Normal Distribution



1. **Bell Shape** – Most values are near the average.
2. **Symmetric** – Equal spread on both sides of the mean.
3. **Mean = Median = Mode** – All are the same in normal distribution.



Skewed Distribution

- A **skewed distribution** means the data is **not balanced** around the average (mean). Instead of being symmetric like a normal distribution, one side is **longer or stretched**.

1 Right-Skewed (Positive Skew)

- The **tail is longer on the right side**.
- Most data values are **small**.
- A few **very large values** pull the tail to the right.

Example:

Income data — most people earn moderate income, but a few people earn very high salaries.

2 Left-Skewed (Negative Skew)

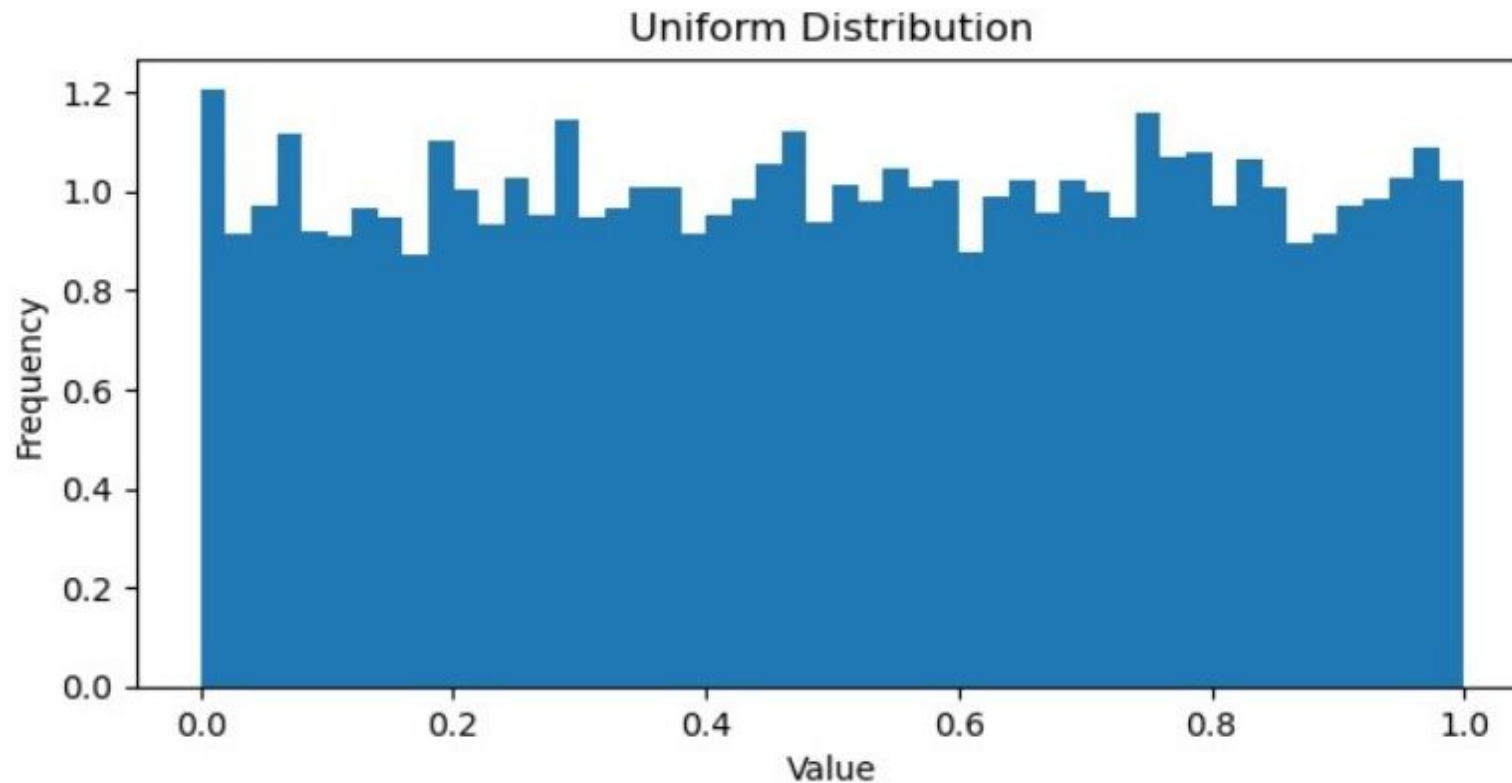
- The **tail is longer on the left side**.
- Most data values are **large**.
- A few **very small values** pull the tail to the left.

Example:

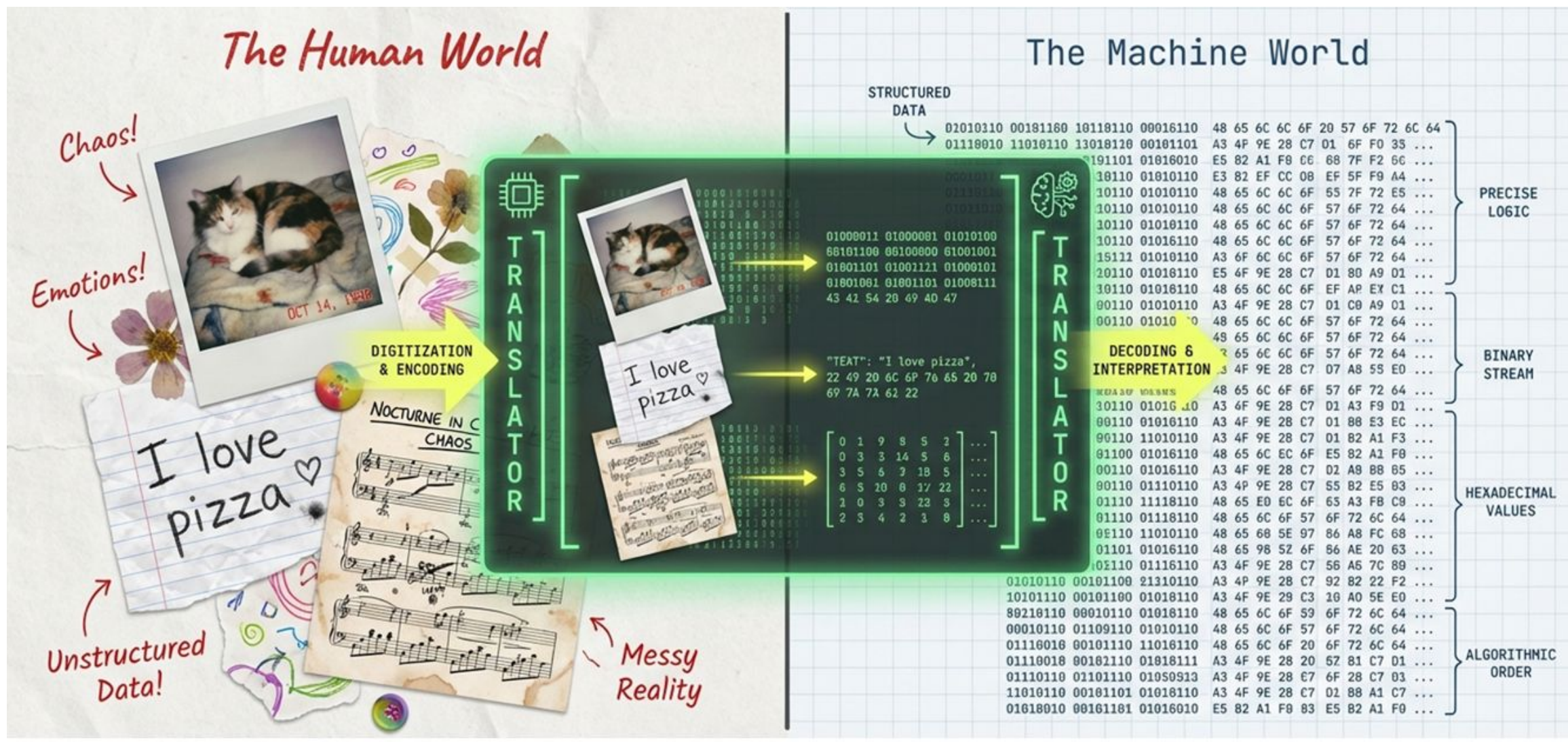
Easy exam scores — most students score high, but a few score very low.

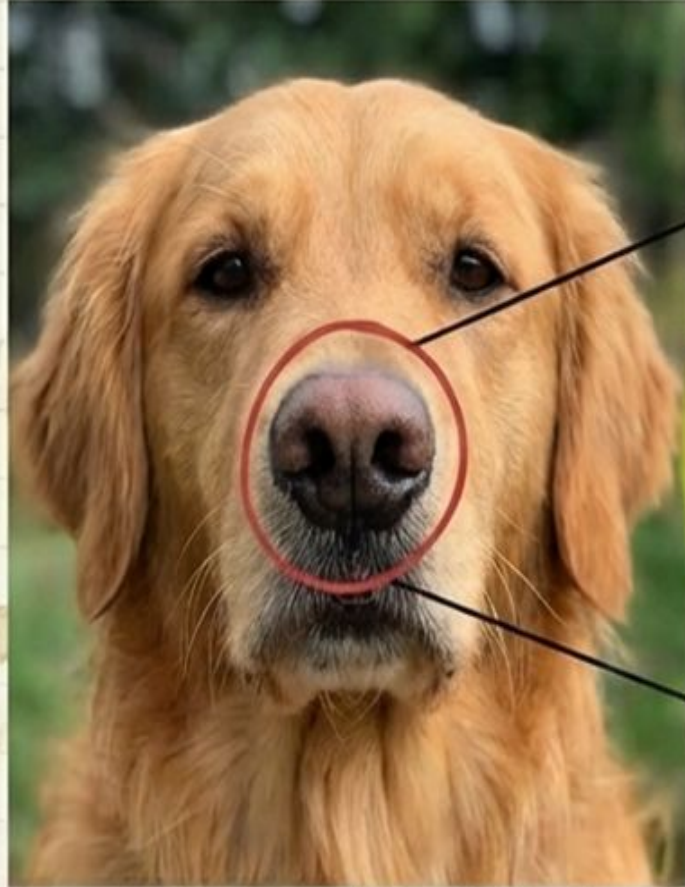
Uniform Distribution

A uniform distribution means **every possible value has the same chance of happening**. There is **no favoritism** toward any number.



Linear algebra (applied focus)

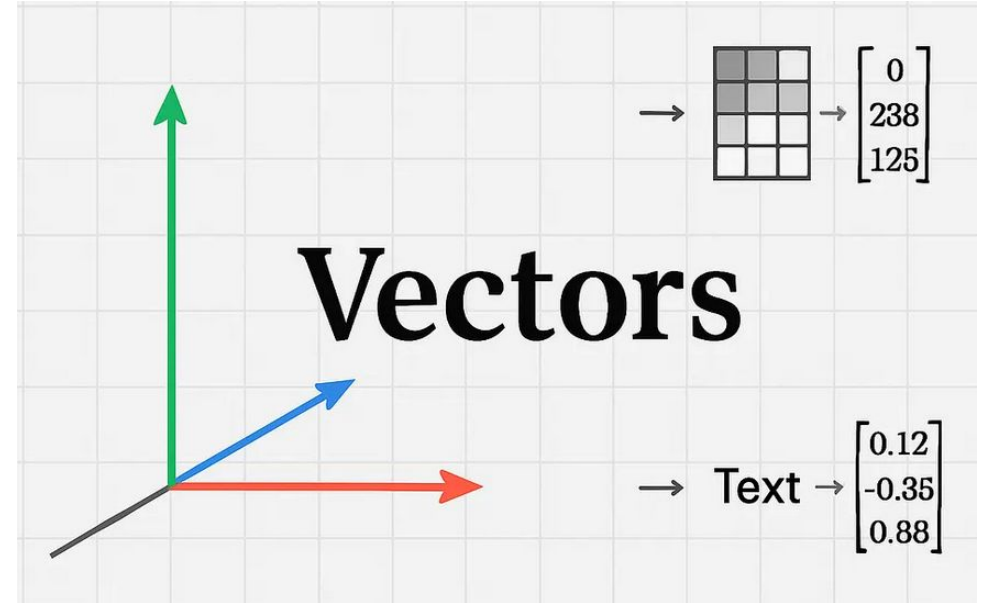




45	200	88	255	12	150	23	78	255
200	88	255	45	12	23	78	<u>95</u>	150
45	200	45	150	88	150	23	78	61
88	88	81	77	255	12	150	28	64
82	250	255	12	12	<u>150</u>	23	24	56
75	63	82	60	78	2	150	12	31
45	200	150	130	255	100	150	23	255
45	255	12	76	14	100	18	81	26
23	65	150	<u>150</u>	100	12	78	23	78
78	45	32	65	255	<u>88</u>	42	87	255

Vector

- Vectors represent data as numbers.
- They have magnitude and direction.
- Each element is a feature.
- Used in classification and clustering.
- Two-dimensional example: $v = (x_1, x_2)$.





Vector Operations

1. Vector Addition and Subtraction
2. Scalar Multiplication
3. Dot Product
4. Cross Product



Numericals

$$A = (3, 5), \quad B = (2, 4)$$

Find $A + B$ and $A - B$.

Find the dot product.

Cross Product

$$A = (a_1, a_2, a_3)$$

$$B = (b_1, b_2, b_3)$$

$$A \times B = (a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1)$$

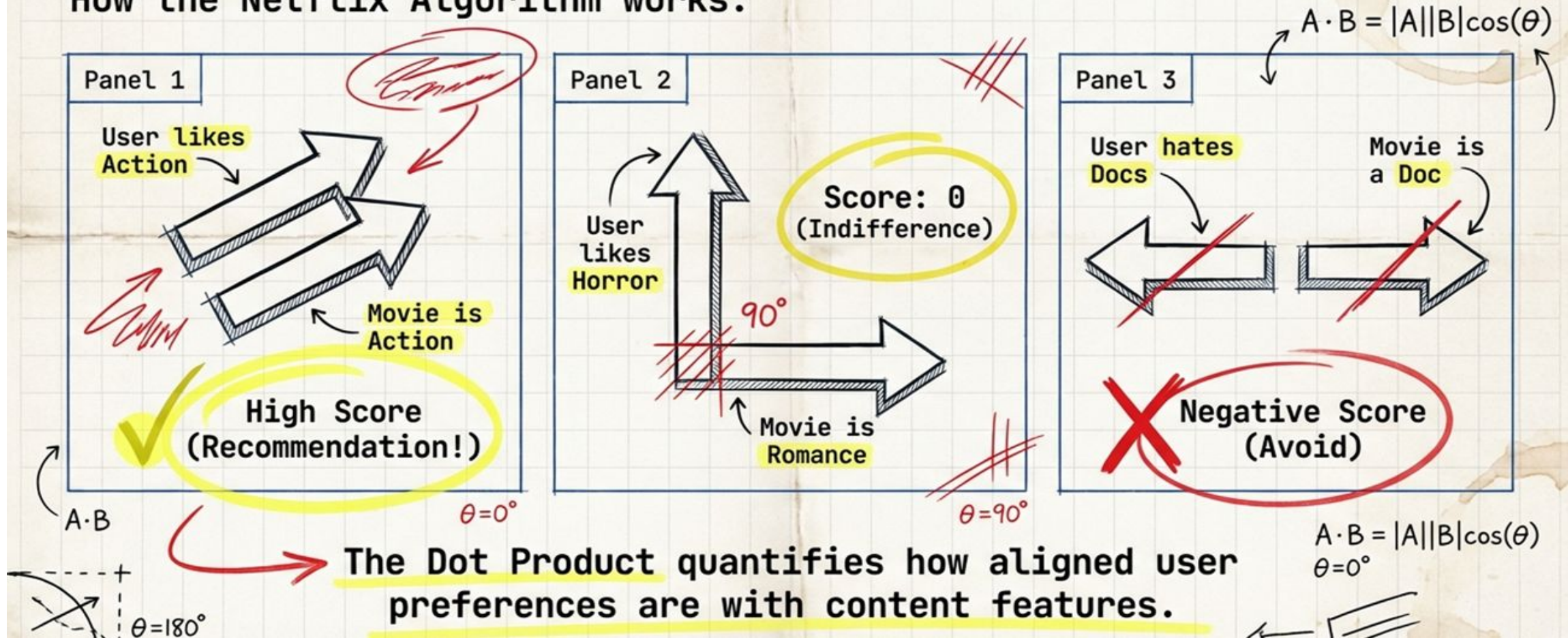
Question

$$A = (1, 2, 3)$$

$$B = (4, 5, 6)$$

Find $A \times B$.

How the Netflix Algorithm works.





Matrix

- A matrix is a table of numbers.
- It represents multiple data points together.
- Rows often represent samples.
- Columns represent features of data.
- Used in training machine learning models.
- Enables efficient computation with vectors.

Suppose we have 3 students with 2 features (Math score, English score):

$$X = \begin{bmatrix} 80 & 90 \\ 70 & 85 \\ 95 & 88 \end{bmatrix}$$

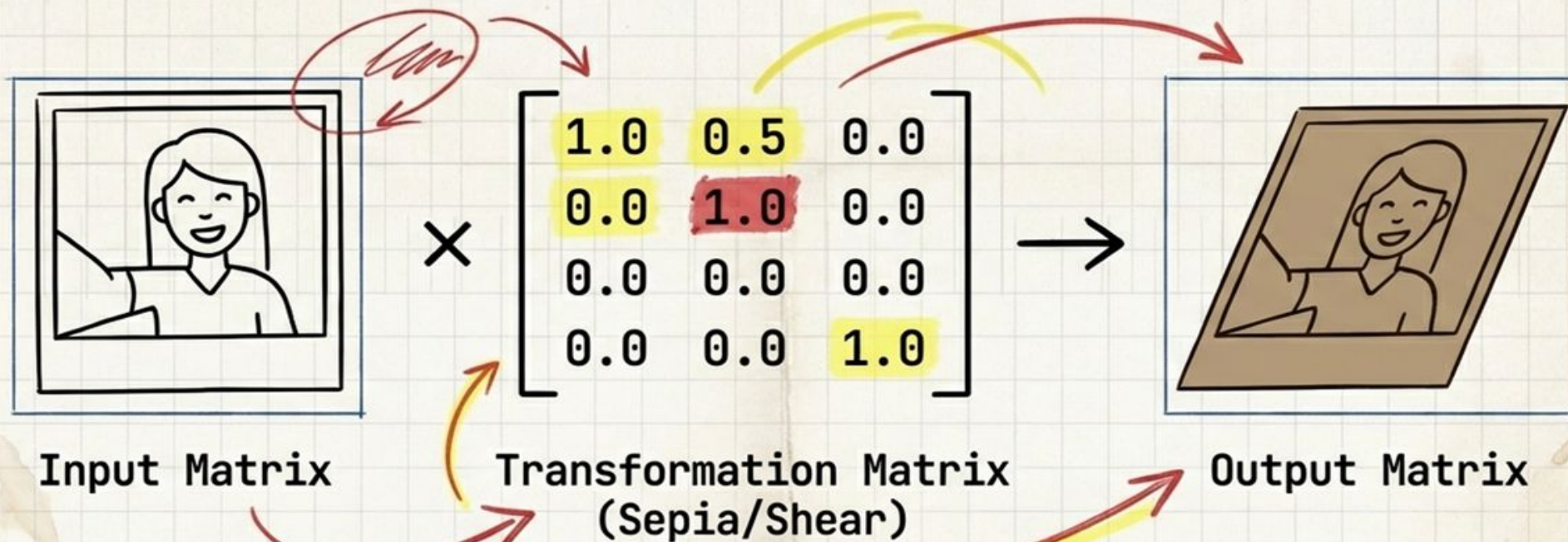
- Rows → each student (sample)
- Columns → features (Math, English)

Apps like Facebook use matrices to:

- Store image pixels
- Compare faces
- Detect patterns

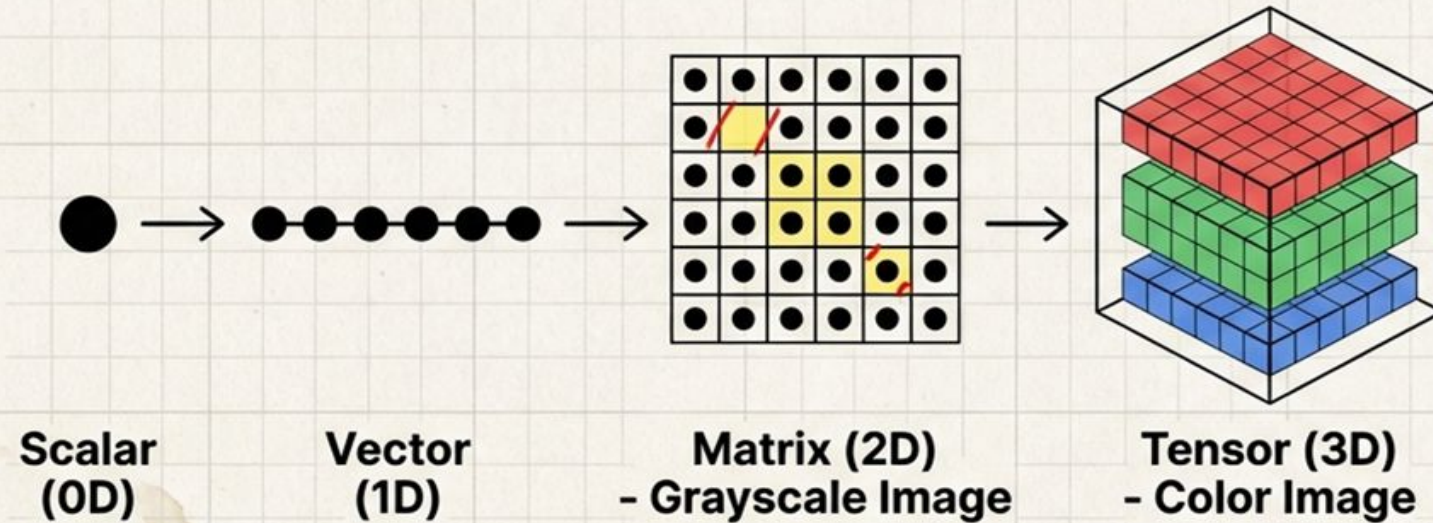
Matrix Multiplication Transforms Reality

Every Instagram filter is just a matrix multiplication that scales, rotates, or skews your pixel data.



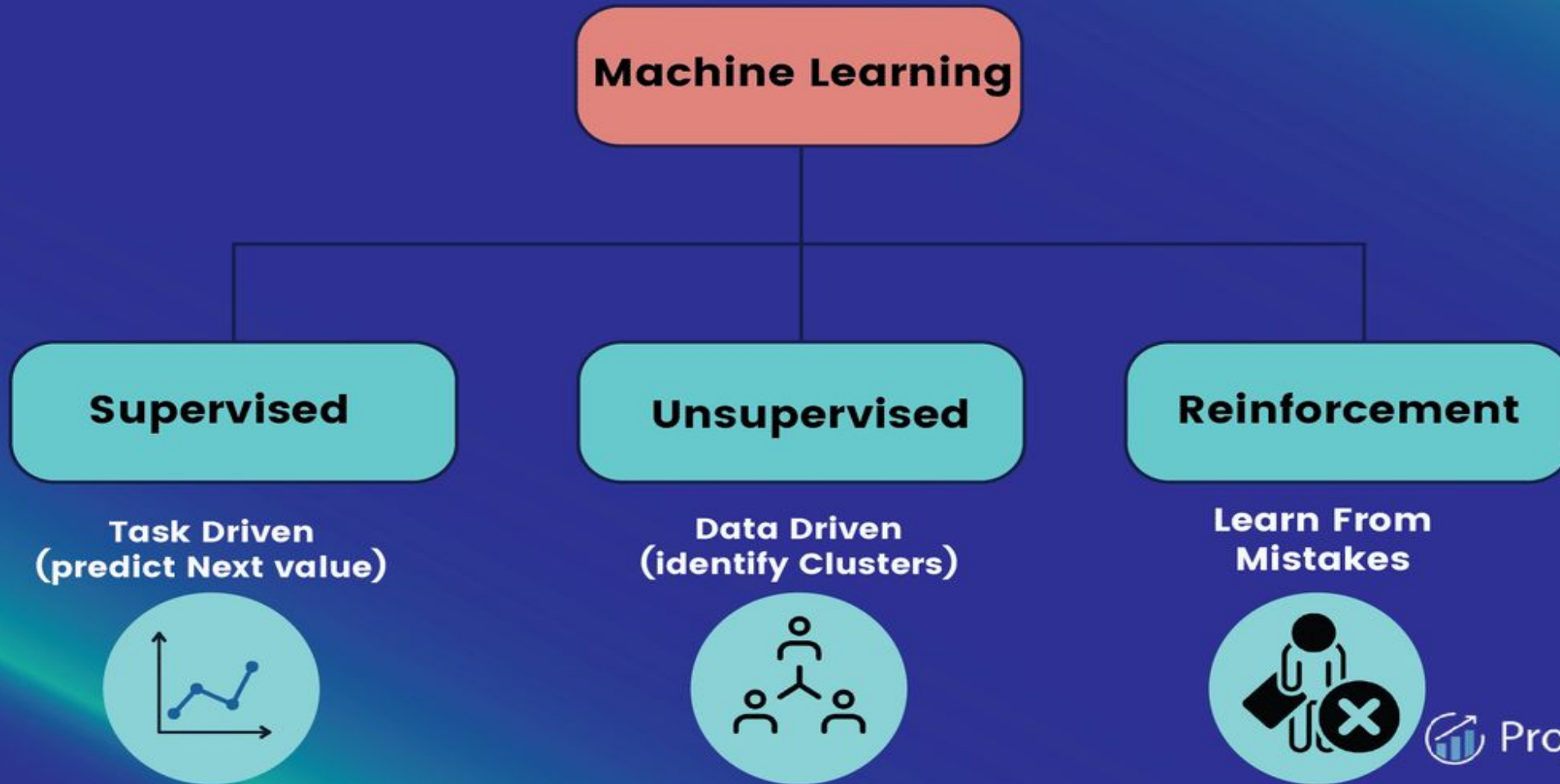
Tensors: The 'T' in TensorFlow

Tensors are just containers for data in n-dimensions.



...Modern AI Toolkits
(TensorFlow, PyTorch)...

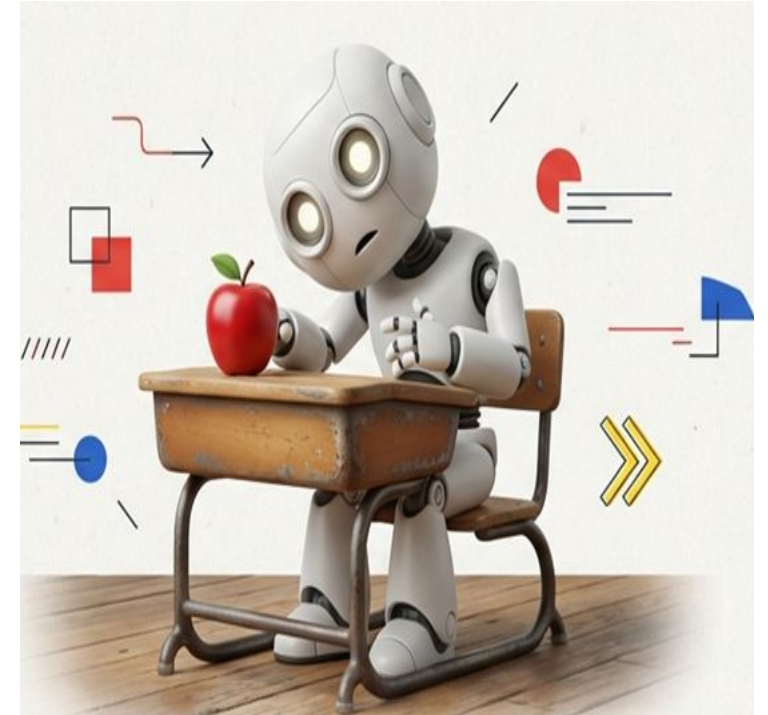
Types of Machine Learning



1. Supervised learning

Supervised Learning is a type of Machine Learning where the computer learns from **labeled data**.

This means the data already has **input and correct output**, and the model learns to predict the output.



INPUT RAW DATA**Labeled Data****Labels**

Supervised Machine Learning

Model is trained using labeled data and pre-processed to predict outcomes and classify data accurately.

OUTPUT**Processing****Algorithm**

Elephant



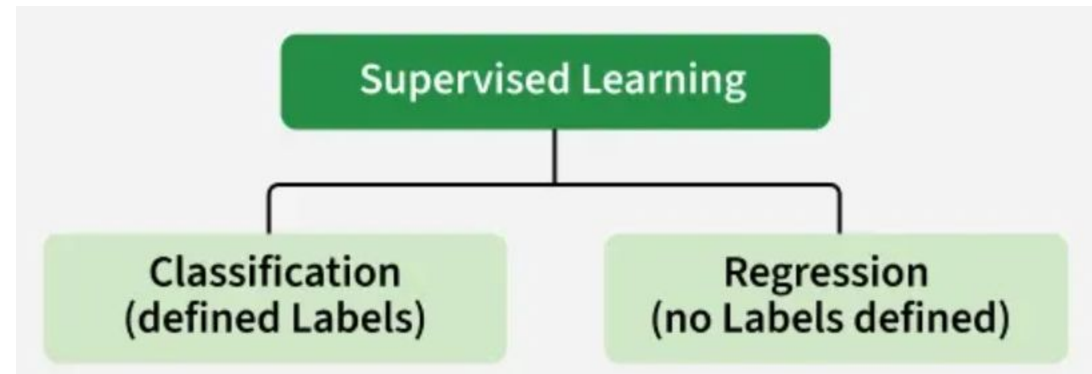
Camel



Cow



Types of Supervised Learning



User ID	Gender	Age	Salary	Purchased	Temperature	Pressure	Relative Humidity	Wind Direction	Wind Speed
15624510	Male	19	19000	0	10.69261758	986.882019	54.19337313	195.7150879	3.278597116
15810944	Male	35	20000	1	13.59184184	987.8729248	48.0648859	189.2951202	2.909167767
15668575	Female	26	43000	0	17.70494885	988.1119385	39.11965597	192.9273834	2.973036289
15603246	Female	27	57000	0	20.95430404	987.8500366	30.66273218	202.0752869	2.965289593
15804002	Male	19	76000	1	22.9278274	987.2833862	26.06723423	210.6589203	2.798230886
15728773	Male	27	58000	1	24.04233986	986.2907104	23.46918024	221.1188507	2.627005816
15598044	Female	27	84000	0	24.41475295	985.2338867	22.25082295	233.7911987	2.448749781
15694829	Female	32	150000	1	23.93361956	984.8914795	22.35178837	244.3504333	2.454271793
15600575	Male	25	33000	1	22.68800023	984.8461304	23.7538641	253.0864716	2.418341875
15727311	Female	35	65000	0	20.56425726	984.8380737	27.07867944	264.5071106	2.318677425
15570769	Female	26	80000	1	17.76400389	985.4262085	33.54900114	280.7827454	2.343950987
15606274	Female	26	52000	0	11.25680746	988.9386597	53.74139903	68.15406036	1.650191426
15746139	Male	20	86000	1	14.37810685	989.6819458	40.70884681	72.62069702	1.553469896
15704987	Male	32	18000	0	18.45114201	990.2960205	30.85038484	71.70604706	1.005017161
15628972	Male	18	82000	0	22.54895853	989.9562988	22.81738811	44.66042709	0.264133632
15697686	Male	29	80000	0	24.23155922	988.796875	19.74790765	318.3214111	0.329656571
15733883	Male	47	25000	1					

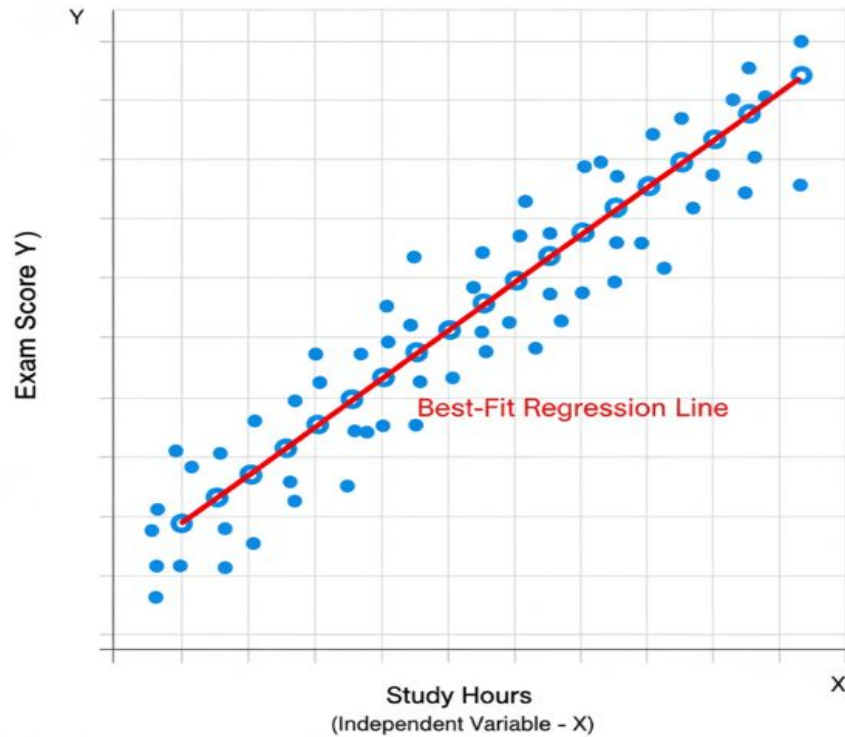
Linear Regression

It is the technique that models the linear relationship between input features (independent variables) and a target value (dependent variable) to make predictions

Square Feet (X)	House Price (Y)
1300	240
1500	320
1700	330
1830	295
1550	256
2350	409
1450	319

Linear Regression

Simple Linear Regression: Study Hours vs. Exam Scores



Where:

- m = slope
- b = intercept

$$Y = mX + b$$



Real Use Cases

- Zillow, Redfin (real estate)
- Stock price prediction
- Demand forecasting
- Climate modeling
- Medical dosage calc

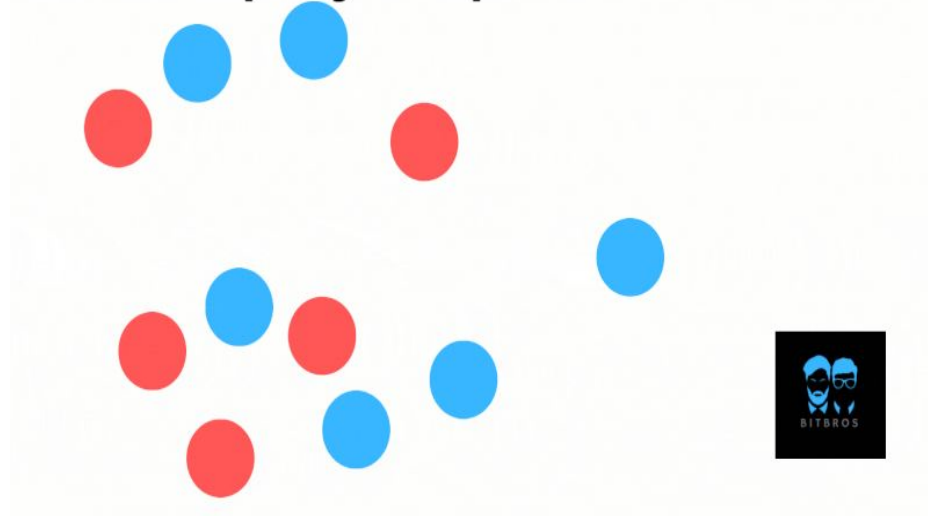
Pros & Cons

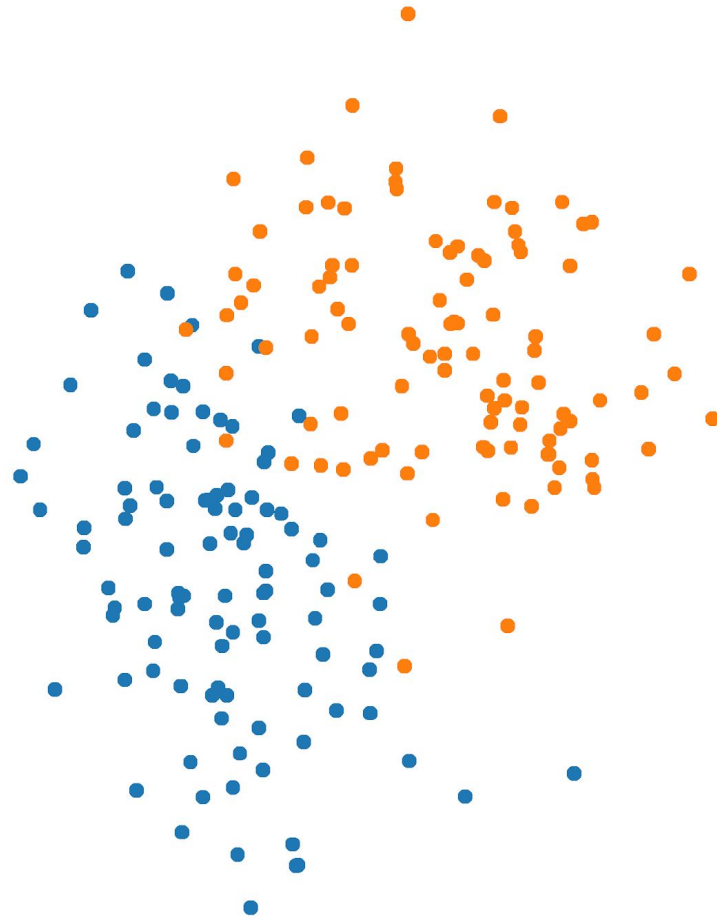
- ✓ Very fast
- ✓ Interpretable
- ✓ Simple baseline
- X Linear only
- X Poor for complex data

KNN

- K-Nearest Neighbours is a supervised learning algorithm.
- It is used for classification and regression tasks.
- It works by finding the nearest data points.
- K decides how many neighbours are used.
- It predicts based on a majority vote or average.

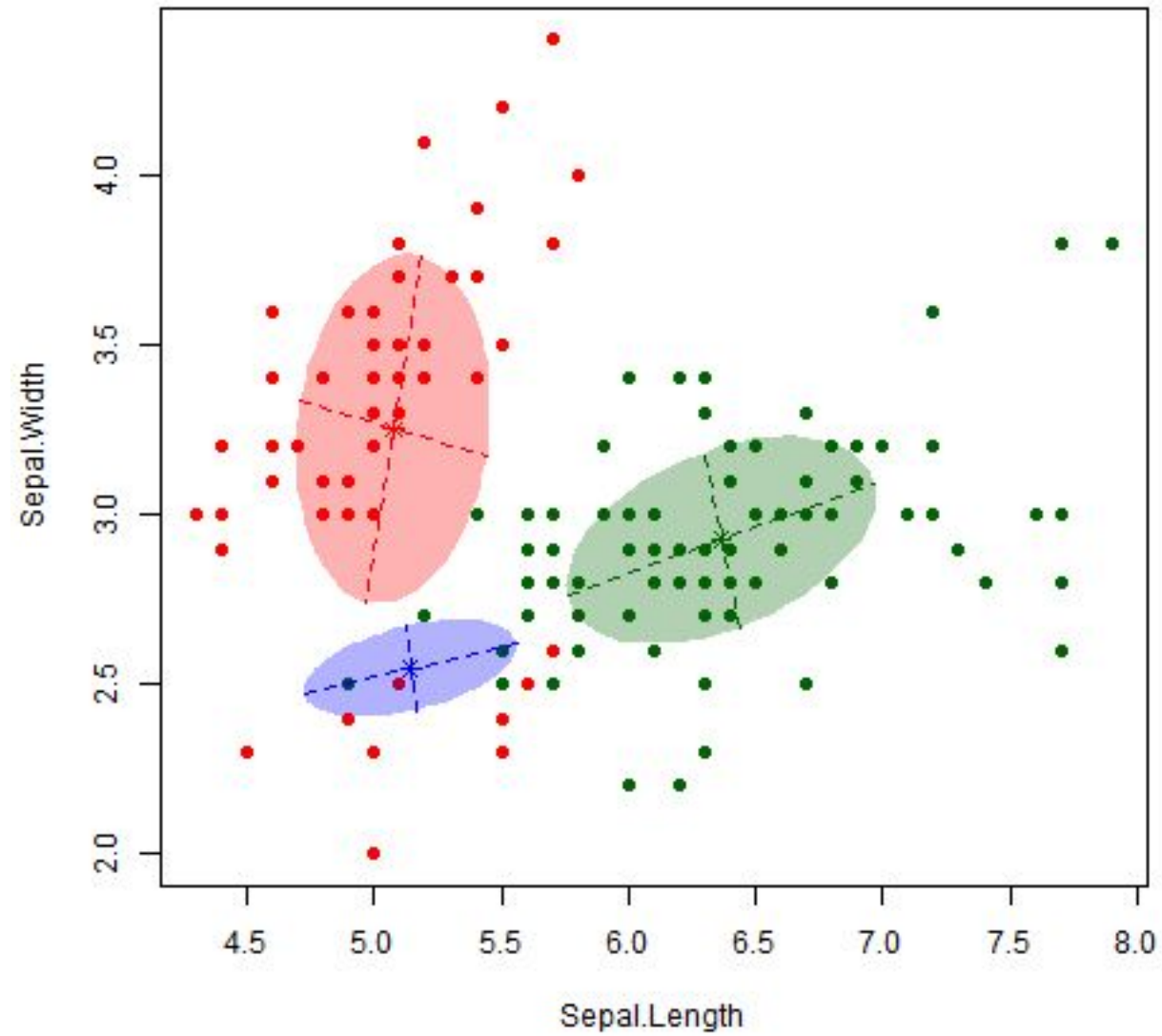
KNN Step-By-Step





a) A labeled dataset has three distinct groups

- Class 1
- Class 2





Real Use Cases

- Netflix recommendations
- Amazon product similarity
- Spotify music discovery
- Anomaly detection
- Image search

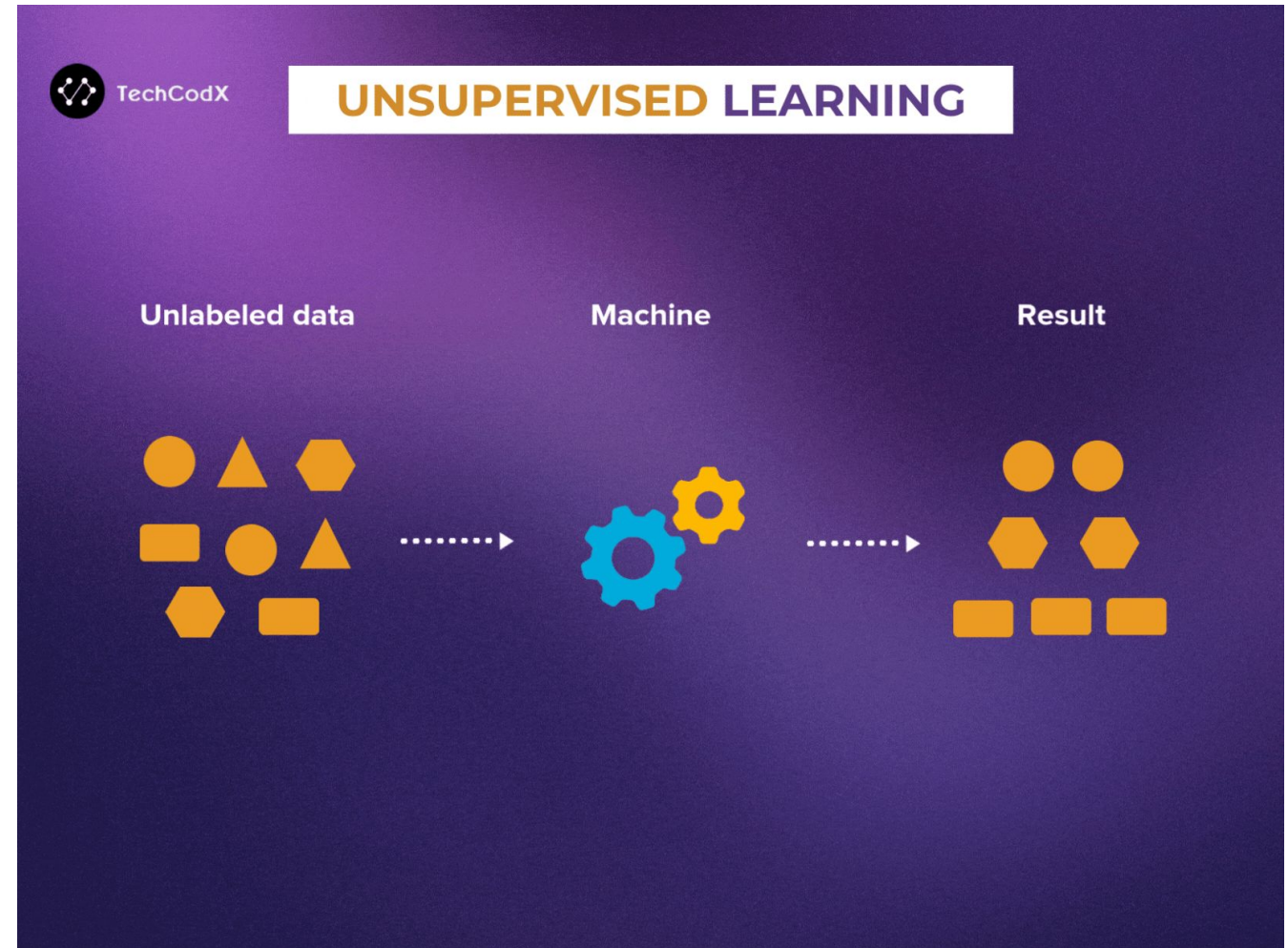
Pros & Cons

- ✓ Simple & intuitive
- ✓ No training needed
- ✓ Works with non-linear

- ✗ Slow prediction
- ✗ Memory intensive

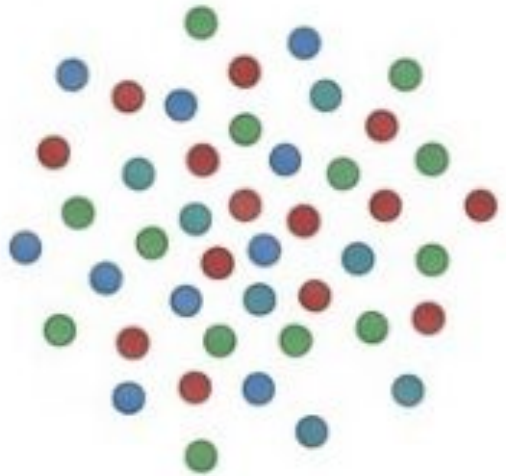
2. Unsupervised learning

Unsupervised learning is a machine learning approach where algorithms analyze and group unlabeled data to discover patterns, similarities, or underlying structures without predefined labels or outcomes.

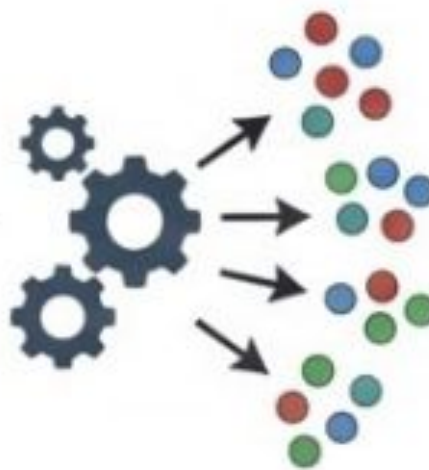


Unsupervised Machine Learning

**1. Input:
Unlabeled Data**



**2. Algorithm
Processing**



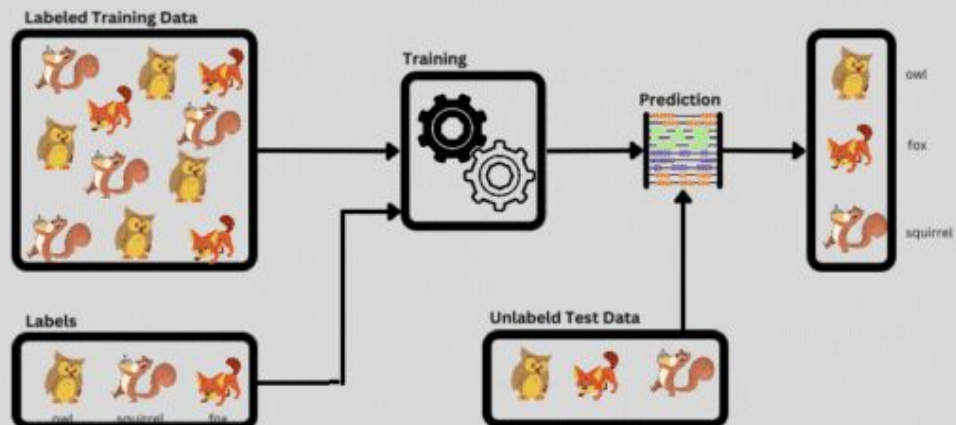
**3. Output:
Clustered Data**



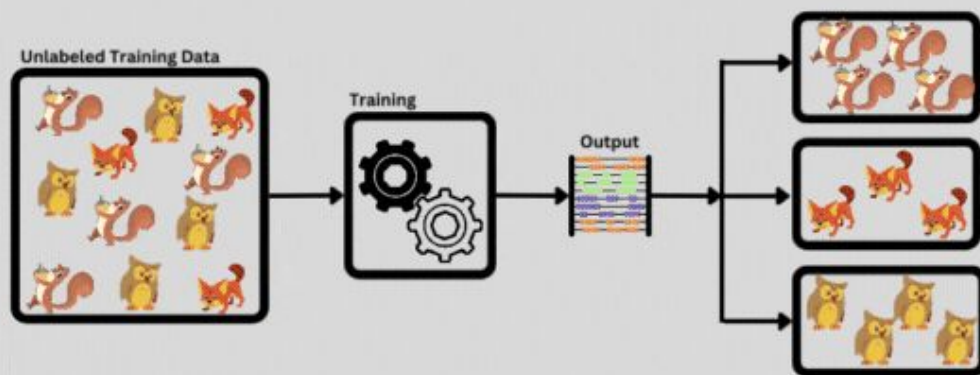
Mechanism: Finding patterns and structures in unlabeled data.

Machine Learning

Supervised Learning



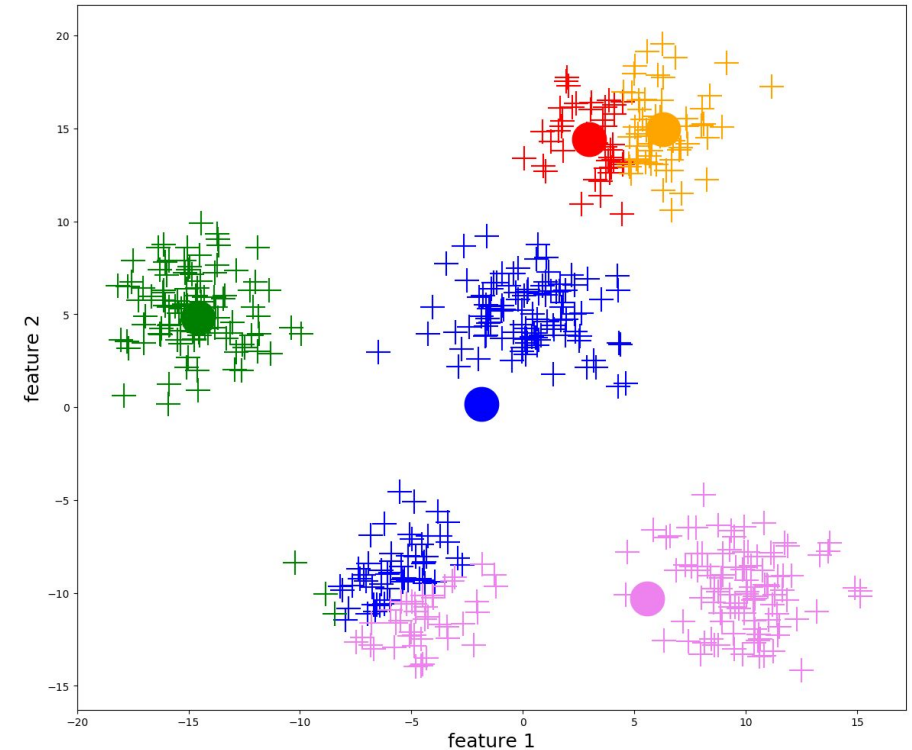
Unsupervised Learning



Datamapu

K-Means clustering

- K-means clusters data into groups.
- Requires a predefined number of clusters.
- Initializes centroids randomly in space.
- Assigns points to the nearest centroid.
- Updates centroids using cluster means.
- Repeats until convergence is achieved.



K-Means Clustering – Solved Example

Initial Centroids:

A1: (2, 10)

B1: (5, 8)

C1: (1, 2)

Data Points			Distance to						Cluster	New Cluster
			2	10	5	8	1	2		
A1	2	10								
A2	2	5								
A3	8	4								
B1	5	8								
B2	7	5								
B3	6	4								
C1	1	2								
C2	4	9								

K-Means Clustering – Solved Example

Current Centroids:

A1: (3.67, 9)

B1: (7, 4.33)

C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			3.67	9	7	4.33	1.5	3.5		
A1	2	10	1.94		7.56		6.52		1	1
A2	2	5	4.33		5.04		1.58		3	3
A3	8	4	6.62		1.05		6.52		2	2
B1	5	8	1.67		4.18		5.70		1	1
B2	7	5	5.21		0.67		5.70		2	2
B3	6	4	5.52		1.05		4.53		2	2
C1	1	2	7.49		6.44		1.58		3	3
C2	4	9	0.33		5.55		6.04		1	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



Real Use Cases

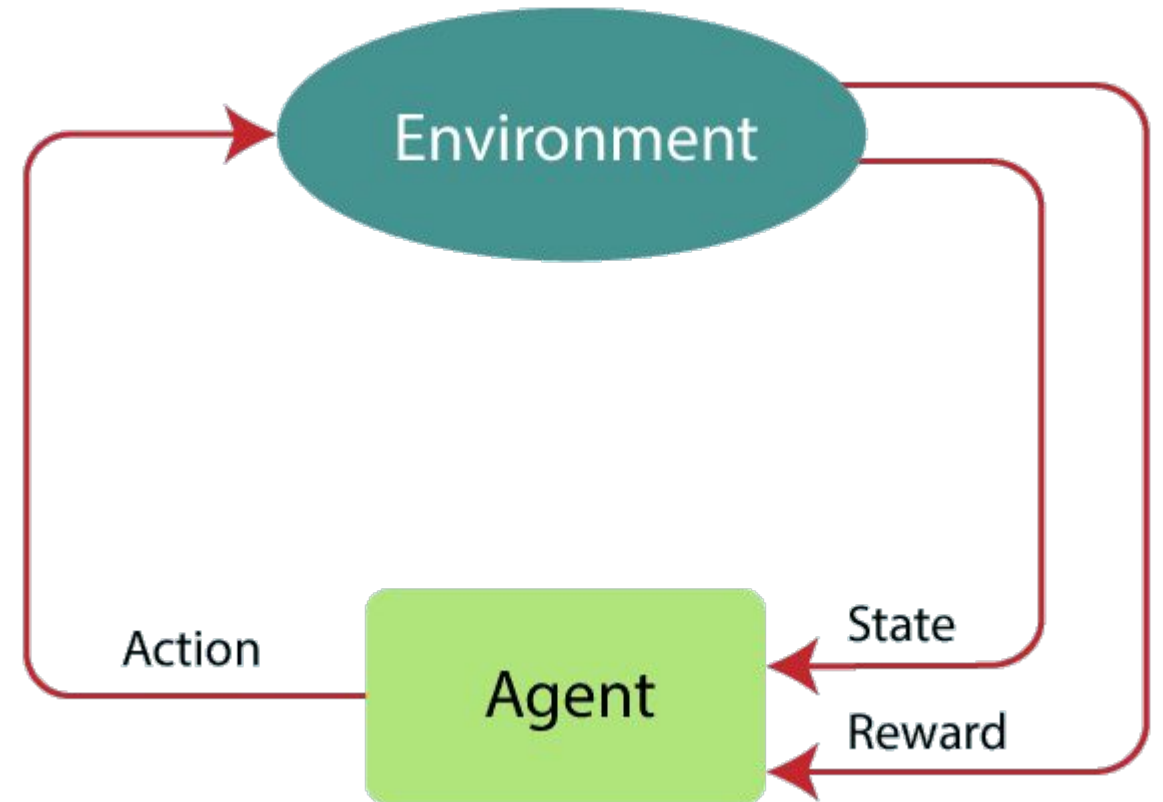
- Customer segmentation
- Image compression
- Document clustering
- Gene sequence analysis
- Recommendation systems

Pros & Cons

- ✓ Fast & scalable
- ✓ Simple to implement
- ✓ Works well for spherical
- ✗ Need to specify k
- ✗ Sensitive to initialization

3. Reinforcement Learning

- **Reinforcement Learning** is a type of machine learning where an **agent** learns by interacting with an **environment** and improves its performance by receiving **rewards** or **penalties**.



Markov Decision Process (MDP)

- It is a mathematical framework used in Reinforcement Learning to model decision-making.
- It describes an environment using **states, actions, transition probabilities, and rewards**.
- The **Markov property** means the next state depends only on the current state and action, not past history.
- An MDP is represented as a tuple (S, A, P, R, γ) .
- The goal in an MDP is to find an optimal policy that maximizes cumulative future reward.

In a **Markov Decision Process (MDP)** represented as (S, A, P, R, γ) :

1.S (States) – The set of all possible situations in which the agent can be.

Example: In a game, each board position is a state.

2.A (Actions) – The set of all possible actions the agent can take in each state.

Example: Move left, move right, jump, etc.

3.P (Transition Probability) – The probability of moving from one state to another after taking an action.

It is written as:

$P(s' | s, a) \rightarrow$ Probability of reaching next state s' from state s after action a .

4.R (Reward Function) – The immediate reward received after taking an action in a state.

It tells the agent whether the action was good or bad.

5. γ (Gamma – Discount Factor) – A value between 0 and 1 that determines the importance of future rewards.

1. γ close to 1 $\rightarrow$ Future rewards are important

2. γ close to 0 $\rightarrow$ Focus mainly on immediate rewards

Total reward (G) = immediate reward + discounted future rewards

$$G = R_1 + \gamma R_2 + \gamma^2 R_3 + \dots$$

- R_1 = reward now
- R_2 = reward next step
- $\gamma^2 R_3$ = reward two steps later, discounted

Model Evaluation

- Model Evaluation in Machine Learning is the process of assessing how well a trained model performs on data, usually unseen or test data, to measure its accuracy, reliability, and generalization ability.
- It tells us how good the model's predictions are.
- Helps compare different models and select the best one.
- Uses **metrics** like accuracy, precision, recall, F1-score, mean squared error, or R^2 depending on the task.
- Often involves **cross-validation** to test the model on multiple splits of the data.
- Prevents problems like **overfitting** or **underfitting**.
- Goal: Ensure the model performs well **not just on training data, but on new, unseen data**.

Confusion Metrics

- A **confusion matrix** is a table used to evaluate the performance of a classification model by comparing predicted values with actual outcomes. It shows correct and incorrect predictions, helping identify where the model makes mistakes.

		Predicted	
		Positive	Negative
Actual	Positive	True positive	False negative
	Negative	False positive	True negative



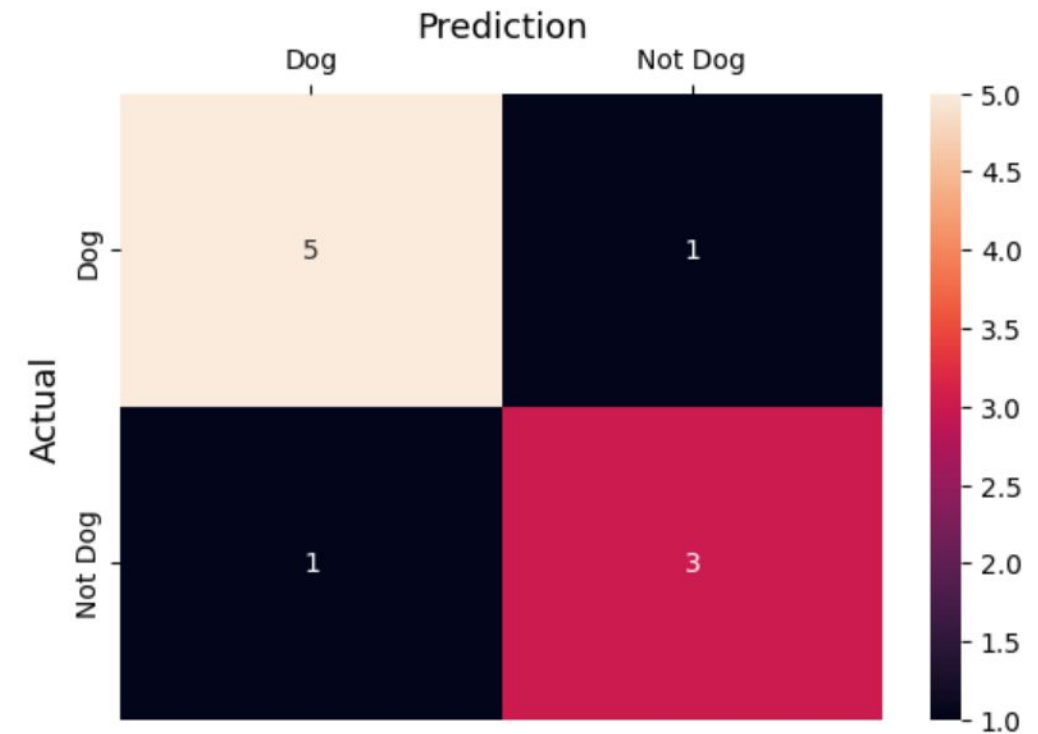
Term	Meaning	Example (Disease Test)
True Positive (TP)	Model predicted positive and it is actually positive	Person has disease and test says "Disease"
False Positive (FP)	Model predicted positive but it is actually negative	Person is healthy but test says "Disease"
False Negative (FN)	Model predicted negative but it is actually positive	Person has disease but test says "No Disease"
True Negative (TN)	Model predicted negative and it is actually negative	Person is healthy and test says "No Disease"



Index	1	2	3	4	5	6	7	8	9	10
Actual	Dog	Dog	Dog	Not Dog	Dog	Not Dog	Dog	Dog	Not Dog	Not Dog
Predicted	Dog	Not Dog	Dog	Not Dog	Dog	Dog	Dog	Dog	Not Dog	Not Dog

Example: Confusion Matrix for Dog Image Recognition with Numbers

Index	1	2	3	4	5	6	7	8	9	10
Actual	Dog	Dog	Dog	Not Dog	Dog	Not Dog	Dog	Dog	Not Dog	Not Dog
Predicted	Dog	Not Dog	Dog	Not Dog	Dog	Dog	Dog	Dog	Not Dog	Not Dog
Result	TP	FN	TP	TN	TP	FP	TP	TP	TN	TN



Visualizing the Confusion Matrix

1. Accuracy : It is the proportion of correct predictions made by the model out of all predictions.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total predictions}}$$

Eg: Imagine a model predicting if an email is spam (1) or not spam (0)

Actual	Predicted
1	1
0	0
1	0
0	0
1	1

2. Precision: Precision measures **how many of the predicted positives are actually positive.**

$$\text{Precision} = \frac{TP}{TP + FP}$$

TP = True Positive (predicted positive & actually positive)

FP = False Positive (predicted positive but actually negative)

Actual	Predicted
1	1
0	1
1	1
0	0
1	0

High precision → fewer false positives.

3. Recall: Recall measures **how many actual positives were correctly predicted.**

$$\text{Recall} = \frac{TP}{TP + FN}$$

TP = True Positive (predicted positive & actually positive)

FP = False Positive (predicted positive but actually negative)

FN = False Negative (actual positive but predicted negative)

Actual	Predicted
1	1
0	1
1	1
0	0
1	0

High recall → model detects most of the actual positive cases.

4. F1 - Score: F1-score is the **harmonic mean of precision and recall**, balancing both:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Introduction to Cross Validation

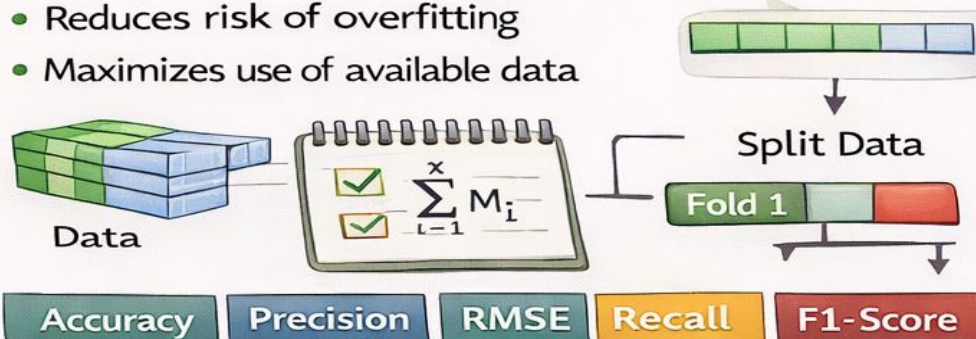
In Machine Learning, we want models to perform well on unseen, new data. A simple approach is to split the data into multiple subsets, training the model on some subsets, and testing it on the remaining one.

Cross Validation

Cross Validation is a technique used to assess how well a model generalizes to **unseen data** by splitting the dataset into multiple subsets, training the model on some subsets, and testing it on the remaining one. This is repeated multiple times.

Why is Cross Validation Needed?

- Better estimate of model performance
- Reduces risk of overfitting
- Maximizes use of available data



Why is Cross Validation Needed?



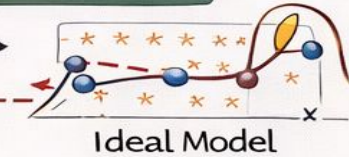
- Better estimate of model performance
- Reduces risk of overfitting



Step 2: Train & Validate Model K Times

- Train on K-1 folds, | K = Bsl $\rightarrow$

• Fold 1 Fold 2 Fold 3 ...



Step 3: Compute CV Score

- CV Score = $\frac{1}{K} \sum M_i \rightarrow M_K$

85% 83% 86% 84.5%

Train, Test, Average

Data Fold 1 Fold 2 Fold 3 Fold 4 Fold 4

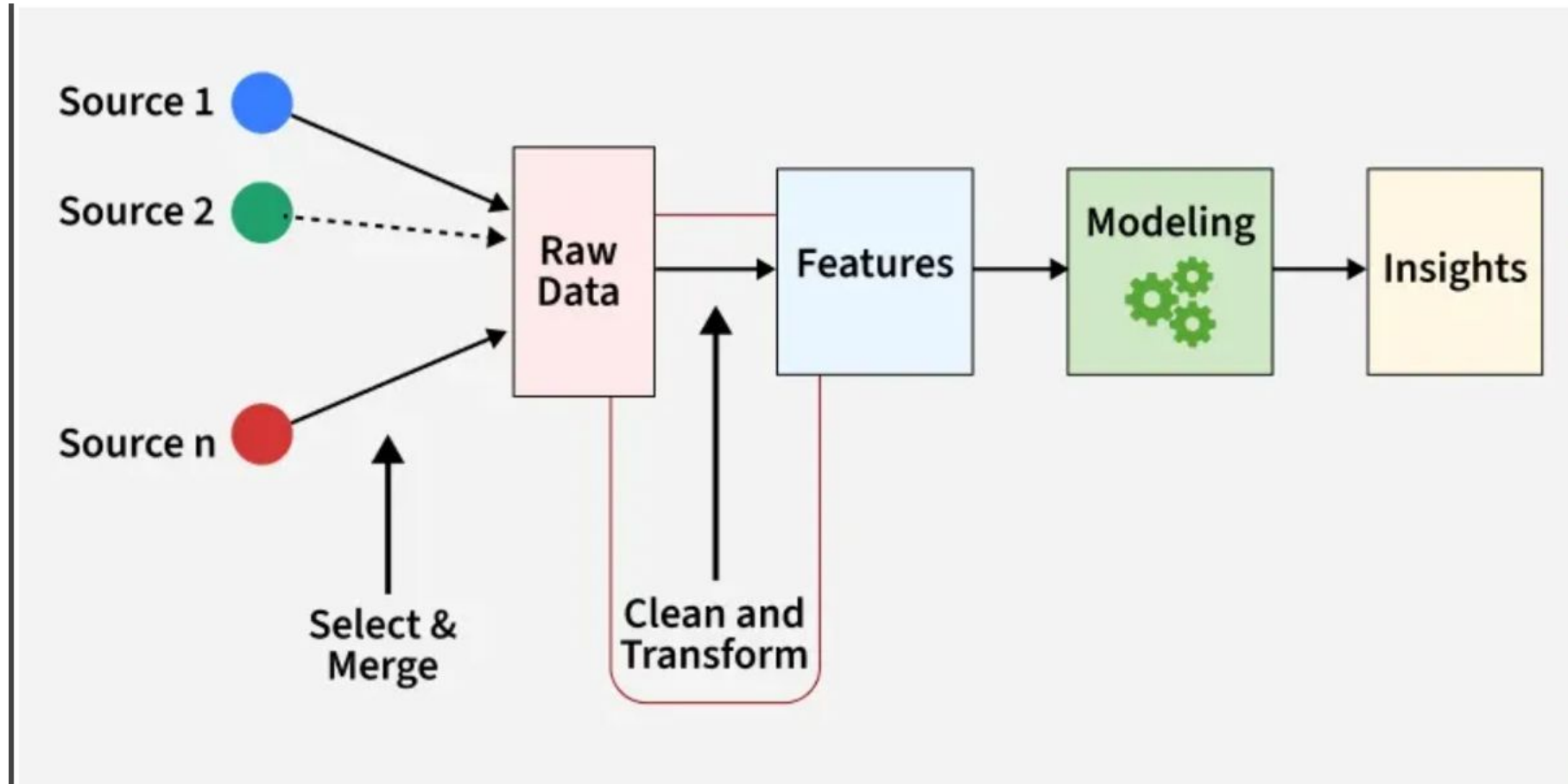
Cross Validation

Technique for testing model performance multiple times and averaging results



Feature engineering

- the process of transforming raw data into meaningful features for machine learning models.
- It involves **creating, selecting, or modifying input variables** to improve model performance.
- Missing values are handled, and categorical variables are encoded into numbers.
- Features are often **scaled or normalized** to ensure consistent ranges.
- New features can be **derived from existing ones** to capture hidden patterns.
- The goal is to provide the model with **informative, relevant, and high-quality data** for better predictions.



Importance

- **Improve accuracy:** Choosing the right features helps the model learn better, leading to more accurate predictions.
- **Reduce overfitting:** Using fewer, more important features helps the model avoid memorizing the data and perform better on new data.
- **Boost interpretability:** Well-chosen features make it easier to understand how the model makes its predictions.
- **Enhance efficiency:** Focusing on key features speeds up the model's training and prediction process, saving time and resources.



Eg: House price prediction

House ID	Location	Bedrooms	Area (sqft)	Age	Has Garage	Price
1	City A	3	1200	10	Yes	250000
2	City B	4	1500	5	No	300000
3	City A	2	800	NaN	Yes	200000
4	City C	3	1000	20	No	180000

Eg: House price prediction

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Step 1: Handle Missing Values

Step 2: Encode Categorical Variables (Feature transformation)

Step 3: Feature Scaling

Step 4: Create New Features

Step 5: Feature Selection

Step 6: Ready for Model

THE COMPREHENSIVE MACHINE LEARNING HIERARCHY

